

LANDSCAPES OF THE CHURCH

06 — Landscape Mosaic & Diversity

Diocesan and Episcopal Conference layers: overview, ArcGIS Online-ready descriptions, methodology, QC, formulas, workflow, and reproducible code

LSAP Integral Ecology Atlas | Technical Documentation | 3 October 2026

11. Analysis 06 — Landscape Mosaic & Diversity

11.1 Diocesan layer overview

Layer name	LandscapesChurch_06_Dioceses_MosaicDiversity_V2
Layer alias	06 - Diocesan Landscape Mosaic & Diversity
Scale / features	Diocese; 2,874 features
Description	Capstone land-cover layer describing compositional diversity, effective land-cover types, leading pair, classified coverage, and descriptive mosaic structure across each diocese.
Attribution	Catholic ecclesiastical geography from the project's versioned diocesan / Episcopal Conference framework; ecclesiastical statistics and leadership enrichment from Holy See statistical publications and Catholic-Hierarchy.org. Environmental land-cover summaries derived from Esri ArcGIS Living Atlas Global Biodiversity and Conservation H3 hexagons.
Citation / credits	Burhans et al., Diocesan Boundaries of the Catholic Church [Feature Layer], v1.0 and subsequent v2.0 project revision; Holy See, Annuario Pontificio / Annuario Statisticum Ecclesiae (multiple years); Catholic-Hierarchy.org (accessed Sep 2026); Esri, Global Biodiversity and Conservation Hexagons (Living Atlas; 2026 release / project-accessed version).
Analyses	Normalized Shannon diversity; effective types; meaningful/major type counts; leading pair/gap; classified coverage; mosaic typology.
Key story fields	STORY_MOSAIC; STORY_STRUCTURE; STORY_SCALE; STORY_CHURCH; STORY_SUMMARY; STORY_COMPREHENSIVE
Status	Publication-ready

11.2 Episcopal Conference layer overview

Layer name	LandscapesChurch_06_Conferences_MosaicDiversity_V2
Layer alias	06 - Episcopal Conference Landscape Mosaic & Diversity
Scale / features	Episcopal Conference; 122 features
Description	Capstone Conference layer summarizing land-cover composition and evenness while integrating Conference leadership, Church statistics, and LSAP participation context.
Attribution	Catholic ecclesiastical geography from the project's versioned diocesan / Episcopal Conference framework; ecclesiastical statistics and leadership enrichment from Holy See statistical publications and Catholic-Hierarchy.org. Environmental land-cover summaries derived from Esri ArcGIS Living Atlas Global Biodiversity and Conservation H3 hexagons.
Citation / credits	Burhans et al., Diocesan Boundaries of the Catholic Church [Feature Layer], v1.0 and subsequent v2.0 project revision; Holy See, Annuario Pontificio / Annuario Statisticum Ecclesiae (multiple years); Catholic-Hierarchy.org (accessed Sep 2026); Esri, Global Biodiversity and Conservation Hexagons (Living Atlas; 2026 release / project-accessed version).
Analyses	Same mosaic/diversity metrics as dioceses plus

	Conference/Church/LSAP storytelling.
Key story fields	Diocesan story set plus STORY_LSAP.
Status	Publication-ready

11.3 Coverage correction

The mosaic analysis explicitly excludes Cloud and NoData from the eight-class diversity calculation. The eight valid class percentages are normalized to their own valid total before calculating Shannon diversity, effective types, and the $\geq 1\%$ / $\geq 5\%$ class counts. TOP_PCT and SECOND_PCT remain percentages of the full geography for intuitive public interpretation.

Coverage fields VALID_COVER_PCT and UNCLASSIFIED_PCT were added after an audit revealed some incomplete source classification, including a small set of diocesan and Conference cases below 80% classified coverage.

11.4 Distribution QC

Retained QC output

06 – MOSAIC & DIVERSITY DISTRIBUTION

```

Dioceses
MOSAIC_DIV:
  10th 27.61 | 25th 39.47 | median 50.71 | 75th 59.17 | 90th 65.12
EFFECTIVE_TYPES:
  10th 1.78 | 25th 2.27 | median 2.87 | 75th 3.42 | 90th 3.87
MEANINGFUL_TYPES (>=1%):
  1:19 | 2:130 | 3:354 | 4:882 | 5:1262 | 6:206 | 7:21
MAJOR_TYPES (>=5%):
  1:126 | 2:689 | 3:1000 | 4:855 | 5:197 | 6:7
TOP_GAP:
  10th 6.03 | 25th 14.66 | median 33.69 | 75th 55.20 | 90th 74.92

Diocesan MOSAIC_CONTEXT:
  Concentrated Landscape: 676 (23.5%)
  Structured Mosaic:      1,346 (46.8%)
  Diverse Mosaic:        715 (24.9%)
  Highly Diverse Mosaic: 137 (4.8%)

Episcopal Conferences
MOSAIC_DIV:
  10th 35.61 | 25th 45.70 | median 54.50 | 75th 62.37 | 90th 66.92
EFFECTIVE_TYPES:
  10th 2.10 | 25th 2.59 | median 3.11 | 75th 3.66 | 90th 4.02
MEANINGFUL_TYPES:
  1:1 | 2:2 | 3:9 | 4:25 | 5:68 | 6:12 | 7:4 | 8:1
MAJOR_TYPES:
  1:3 | 2:29 | 3:38 | 4:37 | 5:15
TOP_GAP:
  10th 5.77 | 25th 15.24 | median 27.81 | 75th 45.12 | 90th 58.88

Conference MOSAIC_CONTEXT:
  Concentrated Landscape: 14 (11.5%)
  Structured Mosaic:      61 (50.0%)
  Diverse Mosaic:        38 (31.1%)
  Highly Diverse Mosaic: 9 (7.4%)

```

11.5 Coverage QC

Retained QC output

CLASSIFIED COVERAGE AUDIT

Dioceses:

```
<95% classified: 57
<90% classified: 22
<80% classified: 3
Minimum:          72.37%
```

Episcopal Conferences:

```
<95% classified: 2
<90% classified: 1
<80% classified: 1
Minimum:          72.24%
```

Interpretation:

Diversity and effective-type calculations describe the CLASSIFIED portion.
Low-coverage cases receive an explicit data note in story text.

11.6 Final publication cleanup QC

Retained QC output

06 – FINAL PUBLICATION CLEANUP

DIOCESES

```
User fields before cleanup: 143
Fields identified for removal: 50
User fields after cleanup: 93
Required publication fields missing: 0
PASS
```

EPISCOPAL CONFERENCES

```
User fields before cleanup: 287
Fields identified for removal: 50
User fields after cleanup: 237
Required publication fields missing: 0
PASS
```

Final map placement:

```
06 - Episcopal Conference Landscape Mosaic & Diversity
06 - Diocesan Landscape Mosaic & Diversity
```

Relevant reproducible code

A.3 Shared land-cover helpers

```
import math

LANDCOVER = {
    "Water": "PCT_LULC_Water",
    "Trees": "PCT_LULC_Trees",
    "Cropland": "PCT_LULC_Crops",
    "Built": "PCT_LULC_Built",
    "Rangeland": "PCT_LULC_Rangeland",
    "Bare": "PCT_LULC_Bare",
    "Snow/Ice": "PCT_LULC_SnowIce",
    "Other Vegetation": "PCT_LULC_Veg"
}

def safe(v):
    return 0.0 if v is None else float(v)

def normalized_classified(raw):
    valid = sum(max(0.0, safe(v)) for v in raw.values())

    if valid <= 0:
        return valid, {k: 0.0 for k in raw}

    return (
        valid,
        {k: 100.0 * max(0.0, safe(v)) / valid for k, v in raw.items()}
    )
```

```

)

def shannon_metrics(raw):
    valid, pct = normalized_classified(raw)

    props = {k: v / 100.0 for k, v in pct.items()}

    H = -sum(
        p * math.log(p)
        for p in props.values()
        if p > 0
    )

    diversity = 100.0 * H / math.log(8)
    effective = math.exp(H)

    meaningful = sum(v >= 1.0 for v in pct.values())
    major = sum(v >= 5.0 for v in pct.values())

    ordered = sorted(
        raw.items(),
        key=lambda kv: safe(kv[1]),
        reverse=True
    )

    top_type, top_pct = ordered[0]
    second_type, second_pct = ordered[1]

    classified_gap = (
        pct[top_type] -
        pct[second_type]
    )

    return {
        "valid_cover": valid,
        "diversity": diversity,
        "effective": effective,
        "meaningful": meaningful,
        "major": major,
        "top_type": top_type,
        "top_pct": safe(top_pct),
        "second_type": second_type,
        "second_pct": safe(second_pct),
        "classified_gap": classified_gap
    }

def mosaic_context(m):
    if (
        m["diversity"] >= 65 and
        m["effective"] >= 4.0 and
        m["meaningful"] >= 5 and
        m["major"] >= 4 and
        m["classified_gap"] <= 15
    ):
        return "Highly Diverse Mosaic"

    if (
        m["diversity"] >= 55 and
        m["effective"] >= 3.25 and
        m["meaningful"] >= 4 and
        m["major"] >= 3 and
        m["classified_gap"] <= 35
    ):
        return "Diverse Mosaic"

    if (
        m["diversity"] < 35 or
        m["effective"] < 2.0 or
        m["major"] <= 1 or
        m["classified_gap"] >= 60
    ):
        return "Concentrated Landscape"

    return "Structured Mosaic"

```

A.9 Analysis 06 — Mosaic & Diversity calculation

```
raw = {
    "Water": safe(row["PCT_LULC_Water"]),
    "Trees": safe(row["PCT_LULC_Trees"]),
    "Cropland": safe(row["PCT_LULC_Crops"]),
    "Built": safe(row["PCT_LULC_Built"]),
    "Rangeland": safe(row["PCT_LULC_Rangeland"]),
    "Bare": safe(row["PCT_LULC_Bare"]),
    "Snow/Ice": safe(row["PCT_LULC_SnowIce"]),
    "Other Vegetation": safe(row["PCT_LULC_Veg"])
}

m = shannon_metrics(raw)

VALID_COVER_PCT = m["valid_cover"]
UNCLASSIFIED_PCT = max(0.0, 100.0 - VALID_COVER_PCT)

MOSAIC_DIV = m["diversity"]
EFFECTIVE_TYPES = m["effective"]
MEANINGFUL_TYPES = m["meaningful"]
MAJOR_TYPES = m["major"]

TOP_TYPE = m["top_type"]
TOP_PCT = m["top_pct"]
SECOND_TYPE = m["second_type"]
SECOND_PCT = m["second_pct"]

TOP_GAP = (
    TOP_PCT -
    SECOND_PCT
)

TOP_GAP_CLASSIFIED = m["classified_gap"]

MOSAIC_CONTEXT = mosaic_context(m)
MOSAIC_PAIR = f"{TOP_TYPE} + {SECOND_TYPE}"
```

A.10 Safe m² → km² conversion

```
import statistics

values = []

with arcpy.da.SearchCursor(
    feature_class,
    ["ANALYSIS_AREA"]
) as cursor:

    for (value,) in cursor:
        if value is not None and value > 0:
            values.append(float(value))

median_area = statistics.median(values)

if median_area >= 1_000_000:

    with arcpy.da.UpdateCursor(
        feature_class,
        ["ANALYSIS_AREA"]
    ) as cursor:

        for row in cursor:
            if row[0] is not None:
                row[0] = float(row[0]) / 1_000_000.0
                cursor.updateRow(row)

else:
    print("Area already appears to be in km²; no conversion performed.")
```

A.16 Publication cleanup pattern

```
# Keep only fields intentionally required for the publication derivative.
```

```

KEEP_FIELDS = {
    "BUFFER_ID",
    "Account_Name",
    "Country",
    "Language",
    "Official_website",
    "Sector",
    "BUFFER_AREA",
    "LANDSCAPE_SCALE",
    # ... eight PCT_LULC fields ...
    # ... coverage/mosaic/transition fields ...
    # ... story fields ...
}

delete_fields = [
    f.name
    for f in arcpy.ListFields(feature_class)
    if (
        not f.required and
        f.name not in KEEP_FIELDS
    )
]

if delete_fields:
    arcpy.management.DeleteField(
        feature_class,
        delete_fields
    )

# Always run a final required-field check after deletion.

```

Sources and attribution

Ecclesiastical geography and Church statistics: LSAP Integral Ecology Atlas / project Catholic jurisdictional framework; Holy See statistical publications; Catholic-Hierarchy.org. Environmental context: Esri ArcGIS Living Atlas Global Biodiversity and Conservation Hexagons and the underlying credited source organizations.

Esri Global Biodiversity and Conservation Hexagons: <https://www.arcgis.com/home/item.html?id=f1bdbdcac3c54ad8b5b21f5b17d91e8d>

Esri 2026 update: <https://www.esri.com/arcgis-blog/products/arcgis-living-atlas/sustainable-development/global-biodiversity-and-conservation-hexagons-but-better>

Summarize Within documentation: <https://pro.arcgis.com/en/pro-app/3.6/tool-reference/analysis/summarize-within.htm>